Nakul Rampal

Website — Email — Phone — X @rampalnakul — LinkedIn — My GitHub — Hugging Face

Professional Summary

I am currently a BIDMaP Fellow at the University of California Berkeley (www.bidmap.berkeley.edu). I work at the intersection of AI and chemistry/materials science, developing LLMs, AI agents, agent harnesses, alignment methods, datasets, benchmarks, and evaluation frameworks for scientific discovery. I also have experience building self-driving labs (automated synthesis + characterization) and integrating AI with the physical world.

Education

University of Cambridge — Cambridge, UK
Ph.D. in Chemical Engineering
Advisor: Prof. David Fairen-Jimenez, Cambridge Trust Scholar — Sep 2019 – Jul 2023

University of California, Berkeley — Berkeley, CA
M.S. in Chemical Engineering; GPA: 3.976/4
Advisor: Prof. Berend Smit, Merit-based Scholarship — Aug 2016 – May 2017

Manipal Institute of Technology — Manipal, India
B.Tech. in Chemical Engineering; GPA: 8.91/10
Thesis with Prof. Ateeque Malani (IIT Bombay) — Jul 2011 – May 2015

Experience

University of California, Berkeley — Berkeley, CA
Postdoctoral Scholar; BIDMaP Fellow
Advisors: Prof. Omar Yaghi, Dean Jennifer Chayes, Prof. Christian Borgs — Jun 2023 – Present

University of California, Berkeley — Berkeley, CA (Hybrid)
Researcher
Advisor: Prof. Berend Smit — Jun 2017 – Jun 2019

Technical Expertise

LLM & Post-Training Tooling: PyTorch, Hugging Face Transformers, Datasets, TRL, PEFT, tokenizers, bitsandbytes, DeepSpeed, Accelerate

Alignment & Evaluation Tooling: lm-eval-harness, OpenAI Evals, Weights & Biases, Humanloop-style evaluation workflows, LLM-as-judge pipelines, preference-data pipelines

Agents & Tool Use: LangChain, LangGraph, Model Context Protocol (MCP), MCP clients/servers, JSON-RPC, OpenAI/Anthropic tool calling, structured outputs, retrieval-augmented generation

Serving & ML Infrastructure: vLLM, FastAPI, Docker, Ray, GitHub Actions, Git/GitHub, Bash, Jupyter, NumPy, pandas, SciPy, scikit-learn

Self-Driving Labs & Automation: Bayesian optimization, closed-loop experimentation, Python instrument control, instrument APIs, data acquisition, LIMS/ELN metadata, protocol standardization

Scientific AI & Materials Tooling: RDKit, pymatgen, ASE, CSD/Python API, RASPA (GCMC), LAMMPS (MD), VASP (DFT), BETSI

Publications

(§ denotes equal contribution, * denotes corresponding author)

1. Synthesis of Highly Crystalline Covalent Organic Frameworks Using Large Language Models
   Kaiyu Wang, Daehyun Daniel Ahn, **Nakul Rampal**, . . . , Jennifer T. Chayes, Omar M. Yaghi*
   Journal of the American Chemical Society, 2026. C&EN
   Featured in C&EN.

2. An Automated Evaluation Agent for Q&A Pairs and Reticular Synthesis Conditions
   Nakul Rampal[§], Dongrong Joe Fu[§], Chengbin Zhao, Christian Borgs*, Jennifer T. Chayes*, Omar M. Yaghi*
   Digital Discovery, 2026. GitHub

3. Reticular Chemistry: Past, Present, and Future
   Haozhe Li, **Nakul Rampal**, Omar M. Yaghi*
   Molecular Frontiers Journal, 2025. Cover
   Featured on the cover of Molecular Frontiers Journal.

4. Large language models for reticular chemistry
   Zhiling Zheng, **Nakul Rampal**, Théo Jaffrelot Inizan, Christian Borgs, Jennifer Chayes, Omar Yaghi*
   Nature Reviews Materials, 2025. Cover
   Featured on the cover of Nature Reviews Materials (Volume 10 Issue 5).

5. Comparison of LLMs in Extracting Synthesis Conditions and Generating Q&A Datasets for Metal-Organic Frameworks
   Yuang Shi[§], **Nakul Rampal**[§], Chengbin Zhao, . . . , Jennifer T. Chayes, Omar Yaghi*
   Digital Discovery, 2025. GitHub

6. Single and Multi-Hop Question-Answering Datasets for Reticular Chemistry with GPT-4-Turbo
   Nakul Rampal, Kaiyu Wang, Matthew Burigana, . . . , Christian Borgs*, Jennifer Chayes*, Omar M. Yaghi*
   Journal of Chemical Theory and Computation, 2024. Hugging Face GitHub

7. Better together: Monolithic halide perovskite@metal-organic framework composites
   Elena Avila[§], Hayden Salway[§], Edoardo Ruggeri, . . . , **Nakul Rampal**, . . . , David Fairen-Jimenez*, Miguel Anaya*
   Matter, 2024.

8. Image and Data Mining in Reticular Chemistry Powered by GPT-4V
   Zhiling Zheng, Zhiguo He, Omar Khattab, **Nakul Rampal**, . . . , Jennifer Chayes, Omar Yaghi*
   Digital Discovery, 2024.

9. Harvesting Water from Air with High-Capacity, Stable Furan-Based Metal-Organic Frameworks
   Ali Al-Awadhi, Saumil Chheda, Gautam Stroscio, . . . , **Nakul Rampal**, . . . , Laura Gagliardi, Omar Yaghi*
   Journal of the American Chemical Society, 2024.

10. Shaping the Water-Harvesting Behaviour of Metal-Organic Frameworks Aided by Fine-Tuned GPT Models
    Zhiling Zheng, Ali Al-Awadhi, Saumil Chheda, . . . , **Nakul Rampal**, . . . , Jennifer Chayes, Omar Yaghi*
    Journal of the American Chemical Society, 2023. GitHub Cover
    Featured on the cover of JACS (Volume 145 Issue 51).

11. A GPT-4 Reticular Chemist for Guiding MOF Discovery
    Zhiling Zheng[§], Zichao Rong[§], **Nakul Rampal**, Christian Borgs, Jennifer Chayes, Omar Yaghi*
    Angewandte Chemie, 2023. GitHub

12. ChatGPT Research Group for Optimizing the Crystallinity of MOFs and COFs
    Zhiling Zheng, Oufan Zhang, Lac Ha Nguyen, **Nakul Rampal**, . . . , Jennifer Chayes, Omar Yaghi*
    ACS Central Science, 2023. Cover
    Featured on the cover of ACS Central Science (Volume 9 Issue 11).

13. Advances in surface functionalization of next-generation metal-organic frameworks for biomedical applications: design, strategies, and prospects
    Xu Chen*, Sergio Mercado Argandona, Francesca Melle, **Nakul Rampal**, David Fairen-Jimenez*
    Chem, 2023.

14. Monolithic Zirconium-Based Metal–Organic Frameworks for Energy-Efficient Water Adsorption Applications
    Ceren Camur, Robin Babu, Jose A. Suarez del Pino, **Nakul Rampal**, . . . , Daniel Maspoch, David Fairen-Jimenez*
    Advanced Materials, 2023.

15. Homochiral Porous Metal–Organic Polyhedra with Multiple Kinds of Vertices
    Xianhui Tang[§], Chunlong Meng[§], **Nakul Rampal**, . . . , Yong Cui, Yan Liu*
    Journal of the American Chemical Society, 2023.

16. Sol-Gel Processing of a Covalent Organic Framework for the Generation of Hierarchically Porous Monolithic Adsorbents
Mark Carrington[§], **Nakul Rampal**[§], David G Madden, . . . , Karena W Chapman, David Fairen-Jimenez*
Chem, 2022. Cover
Featured on the cover of Chem (Volume 8 Issue 11).

17. Densified HKUST-1 Monoliths as a Route to High Volumetric and Gravimetric Hydrogen Storage Capacity
David G. Madden[§]*, Daniel O'Nolan[§], **Nakul Rampal**[§], . . . , Karena W. Chapman*, David Fairen-Jimenez*
Journal of the American Chemical Society, 2022.

18. How Reproducible Are Surface Areas Calculated from the BET Equation?
Johannes Osterrieth[§], James Rampersad[§], David G. Madden[§], **Nakul Rampal**[§], . . . , Kirchon Angelo, David Fairen-Jimenez*
Advanced Materials, 2022. GitHub Chemistry World Cover
Featured on the inside back cover of Advanced Materials and in Chemistry World.

19. Lanthanide metal–organic frameworks for the fixation of CO_2 under aqueous-rich and mixed-gas conditions
David H. Le, Ryan P. Loughan, Andrzej Gładysiak, **Nakul Rampal**, . . . , David Fairen-Jimenez, Kyriakos C. Stylianou*
Journal of Materials Chemistry A, 2022.

20. Endohedral functionalization of chiral metal-organic cages for encapsulating achiral dyes to induce circularly polarized luminescence
Xianhui Tang, Hong Jiang, Yubing Si, **Nakul Rampal**, . . . , Yong Cui, Yan Liu*
Chem, 2021.

21. The Development of a Comprehensive Toolbox Based on Multi-Level, High-Throughput Screening of MOFs for CO/N_2 Separations
Nakul Rampal[§], Abdulmalik Ajenifuja[§], Andi Tao[§], . . . , Peyman Z. Moghadam, David Fairen-Jimenez*
Chemical Science, 2021.

22. Formulation of Metal–Organic Framework-Based Drug Carriers by Controlled Coordination of Methoxy PEG Phosphate: Boosting Colloidal Stability and Redispersibility
Xu Chen, Yunhui Zhang, **Nakul Rampal**, . . . , Han Yu*, Clare P. Grey, Oren A. Scherman, David Fairen-Jimenez*
Journal of the American Chemical Society, 2021. C&EN Cover
Featured on the cover of JACS (Volume 143 Issue 34) and in C&EN.

23. Monolithic metal-organic frameworks for carbon dioxide separation
David Madden, Robin Babu, Ceren Camur, **Nakul Rampal**, Joaquin Silvestre-Albero, Teresa Curtin, David Fairen-Jimenez*
Faraday Discussions, 2021.

24. Investigation of Adhesion between Heavy Oil/Bitumen and Reservoir Rock: A Molecular Dynamics Study
Bablu Meghwal, **Nakul Rampal**, Ateeque Malani*
Energy & Fuels, 2020.

25. In silico discovery of covalent organic frameworks for carbon capture
Kathryn S. Deeg, Daiane Damasceno Borges, Daniele Ongari, **Nakul Rampal**, . . . , Johanna M. Huck, Berend Smit*
ACS Applied Materials & Interfaces, 2020.

26. Ab Initio Flexible Force Field for Metal-Organic Frameworks Using Dummy Model Coordination Bonds
Sudi Jawahery, **Nakul Rampal**, Seyed Mohamad Moosavi, Mathew Witman, Berend Smit*
Journal of Chemical Theory and Computation, 2019.

27. Structural Behavior of Isolated Asphaltene Molecules at the Oil-Water Interface
Meena B. Singh, **Nakul Rampal**, Ateeque Malani*
Energy & Fuels, 2018.

Preprints

(§ denotes equal contribution, * denotes corresponding author)

1. An Autonomous Passive MOF Water Harvester for Extreme Arabian Desert Conditions
   Ali H. Alawadhi, **Nakul Rampal**, Daniel Ahn, Omar M. Yaghi*
   ChemRxiv, 2026.

2. A chemical language model for reticular materials design
   Dhruv Menon, Vivek Singh, Xu Chen, . . . , **Nakul Rampal**, . . . , Omar M. Yaghi, David Fairen-Jimenez*
   arXiv, 2026. GitHub

3. Fully-stretched, single-crystalline polymers of linear poly[n]catenanes
   Kaiyu Wang, **Nakul Rampal**, Ying Liu, . . . , Yu Han*, Ting Xu*, Omar Yaghi*
   ChemRxiv, 2025.

4. A Multi-Grained Symmetric Differential Equation Model for Learning Protein-Ligand Binding Dynamics
   Shengchao Liu§, Weitao Du§, Yanjing Li, . . . , **Nakul Rampal**, . . . , Hongyu Guo*, Jennifer Chayes*
   arXiv, 2024.

Mentorship

At the University of Cambridge

- Mythili Sutharson, MPhil in Advanced Chemical Engineering, Oct 2019 – Aug 2020; currently Associate Consultant at Bain
- Hiu Ki Wong, Part IIB in Chemical Engineering, Oct 2020 – Apr 2021; currently Data Engineer at Pirical
- George Irving, Part IIB in Chemical Engineering, Oct 2021 – Apr 2022; currently Founding Engineer at Arva AI
- Khalid Al-Otaibi, MPhil in Advanced Chemical Engineering, Apr 2022 – Aug 2022; Saudi Aramco

At UC Berkeley

Current

- Kaiyu Wang, PhD in Chemistry, Sep 2023 – ongoing
- Joe Fu, BS/PhD in Electrical Engineering & Computer Science, Jun 2024 – ongoing
- Daniel Ahn, PhD in Chemistry, Jan 2025 – ongoing
- Joe Zhou, BS in Electrical Engineering & Computer Science, Jan 2026 – ongoing
- Ruikang Wang, BS in Electrical Engineering & Computer Science, Jan 2026 – ongoing

Previous

- Zhiling Zheng, PhD in Chemistry, Jun 2023 – Dec 2023; currently Assistant Professor at Washington University in St. Louis
- Ali Al-Awadhi, PhD in Chemistry, Sep 2023 – Dec 2025; currently Assistant Professor at UAE University
- Juri Al-Johani, BS in Electrical Engineering & Computer Science, Dec 2023 – Jun 2024
- Dongha Kim, BS in Chemical Engineering/Data Science, Jul 2025 – Dec 2025

Awards

- Cambridge International Scholarship
- Trinity Henry-Barlow Scholarship
- Merit-based Scholarship (Fall 2016 and Spring 2017), University of California, Berkeley
- Top 3 of the graduating class, Manipal Institute of Technology
- School Topper Medal, Unified Cyber Olympiad
- School Wiz Kid Medal, 10th National Science Olympiad

Teaching

- Graduate Student Instructor for "Mathematical Methods in Geophysics", Spring 2017, University of California, Berkeley (Nominated for the Outstanding GSI Award)
- Supervisor for the module on "Adsorption and Advanced Nanoporous Materials", Chemical Engineering Tripos, Michaelmas Term 2022, University of Cambridge
- Co-instructor (6-week summer course) on "Adsorption and Nanoporous Materials" run by Cambridge Enterprise — Summer 2021 (1 course), 2022 (2 courses), 2023 (1 course)
- Guest instructor for the course *PHYS H190 Physical Systems by and for Artificial Intelligence*, Spring 2025, University of California, Berkeley
- Guest instructor for *CHEM 96 - Introduction to Research and Study*, College of Chemistry, Fall 2024, University of California, Berkeley

Invited Talks

1. Invited talk at Shanghai Jiao Tong University, 22nd January 2026.
2. Invited talk at IIT Bombay (Department of Chemical Engineering), 9th January 2026.
3. Invited talk at Saudi Aramco R&DC, 6th November 2025.
4. Invited talk at KAUST (Advanced Membranes and Porous Materials Center), 5th November 2025.
5. Invited talk at the University of Milan (Department of Chemistry), 23rd June 2025.
6. Invited talk at The Chemist's Interactions – June Seminar at the University of Milan, 23rd June 2025.
7. Invited talk (virtual/in-person) at Nanyang Technological University (School of Physical and Mathematical Sciences), 21st January 2022.
8. Presentation (virtual) at the 2nd International School on Porous Materials: MOFschool2021, Lake Como School of Advanced Studies, 21–25 June 2021.
9. Presentation (virtual) at the bp-ICAM Annual Conference 2020, 20th–21st October 2020.
10. Invited talk at Chalmers University of Technology (Department of Physics), 4th February 2019.
11. Invited talk at Karlsruhe Institute of Technology (Department of Theoretical Chemical Biology), 1st February 2019.
12. Invited talk at the Theoretical Chemistry Seminar, Norwegian University of Science and Technology (NTNU), 14th December 2018.

References

1. **Prof. Omar Yaghi (2025 Nobel Laureate in Chemistry)**
   University Professor & James and Neeltje Tretter Professor of Chemistry
   Department of Chemistry, UC Berkeley
   Email: yaghi@berkeley.edu
2. **Dean Jennifer Chayes**
   Dean, College of Computing, Data Science and Society
   UC Berkeley
   Email: jchayes@berkeley.edu
3. **Prof. Christian Borgs**
   Director, Bakar Institute of Digital Materials for the Planet
   UC Berkeley
   Email: borgs@berkeley.edu
4. **Prof. David Fairen-Jimenez**
   Professor of Molecular Engineering
   Department of Chemical and Biomolecular Engineering, University of Cambridge
   Email: df334@cam.ac.uk